COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data structure adn analysis	**SECTION:**	CPE21S1
DATE PERFORMED:	14/07/2026	**DATE SUBMITTED:**	14/07/2026
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

ILO A

```cpp
#include <iostream>

int a, b;
int Sumoftwonumbers (int a, int b);
bool testLogicalOperations(bool a, bool b);

int main() {
    // User Input
    std::cout << "Enter first integer: ";
    std::cin >> a;
    std::cout << "Enter second integer: ";
    std::cin >> b;

    int sum = Sumoftwonumbers (a, b);

    // check which integer is greater

    if (a > b){
        std::cout << "\n" << a << " is greater than " << b;

        } else {
        std::cout << "\n" << b << " is greater than " << a << "\n\n";
    }

    bool success = testLogicalOperations(true, false);


    return 0;
}
```

Figure 1. Code to test the functions

```
// Function to find the sum of the two integers
int Sumoftwonumbers (int a, int b) {
    int total = a + b;
    std::cout << "\nThe sum of a and b is equal to " << total;
    return total;
}

bool testLogicalOperations(bool a, bool b) {
    std::cout << "AND operation (a && b): " << (a && b) << std::endl;
    std::cout << "OR operation  (a || b): " << (a || b) << std::endl;
    std::cout << "NOT operation (!a)    : " << (!a) << std::endl;
    std::cout << "NOT operation (!b)    : " << (!b) << std::endl;

    return true;
```

Figure 2: Functions used

```
Output
Enter first integer: 23
Enter second integer: 43

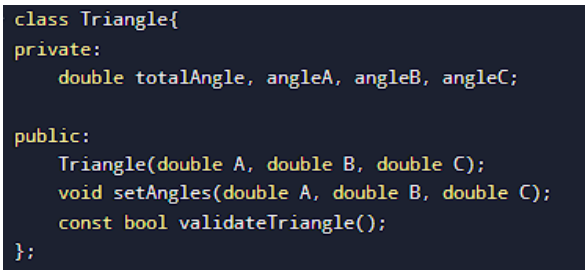
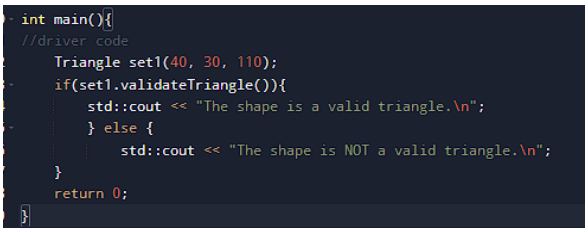
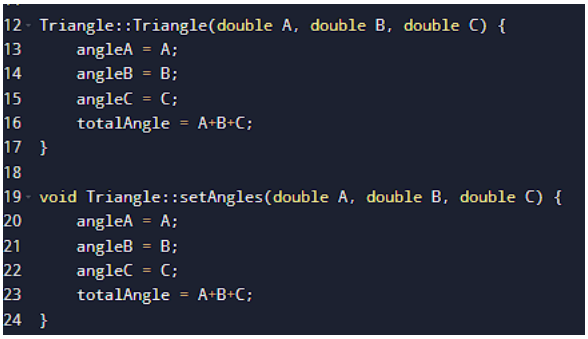
The sum of a and b is equal to 66
43 is greater than 23

AND operation (a && b): 0
OR operation  (a || b): 1
NOT operation (!a)    : 0
NOT operation (!b)    : 1
```

Figure 3: Output of the code

ILO B

Sections	Answer
Header File Declaration Section	`1 #include <iostream>`
Global Declaration Section	

Class Declaration and Method Definition Section	
Main Function	
Method Definition	

```
class Triangle{
private:
    double totalAngle, angleA, angleB, angleC;

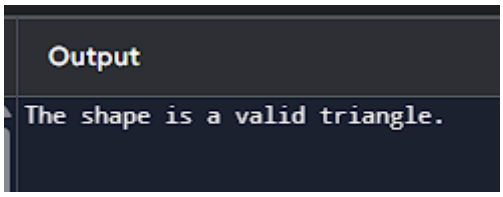
public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();
};
```

```
int main(){
 //driver code
    Triangle set1(40, 30, 110);
    if(set1.validateTriangle()){
        std::cout << "The shape is a valid triangle.\n";
        } else {
            std::cout << "The shape is NOT a valid triangle.\n";
    }
    return 0;
}
```

```
Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
```

Table 1-1 Structure code



Storing totalAngle is a little unnecessary, your private section hides your variables to preserve data integrity. While your public functions contain redundant logic between the constructor and the setter, they do allow external code to interact with the object. Lastly, making validateTriangle() a const member function would improve security by preventing it from changing your personal information.

Table 1-2 Output observations and comments

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1.Create a C++ program to swap the two numbers in different variables.

- I had to think about the logic that went through exchanging the variables, where at first I didn't know what to start with so I did some trial and error until I eventually got this result.

```cpp
#include <iostream>
int temp, tempo;
int a, b;

int main (){
    std::cout << "Enter first number: ";
    std::cin >> a;
    std::cout << "Enter Second number: ";
    std::cin >> b;
    std::cout << "Swapping the numbers....\n";

    temp = a;
    tempo = b;
    a = tempo;
    b = temp;

    std::cout << a << " swapped " << b;

    return 0;
}

```

```
Enter first number: 334
Enter Second number: 542
Swapping the numbers....
542 swapped 334

=== Code Execution Successful ===
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.
   - This was the easiest by far and I even added a lot more functions to the program. I added the function to pick an option between either Fahrenheit to Kelvin or Kelvin to Fahrenheit.

```
#include <iostream>

char choice;
float Kelv, Fahr;
float KelvtoFahr (float Kelv, float Fahr);
float FahrtoKelv (float Kelv, float Fahr);

int main() {

    std::cout << "Welcome to temperature calculator \n";
    std::cout << "----------------------------------\n";
    std::cout << "Kelvin -> Fahrenheit [F]\n";
    std::cout << "Fahrenheit -> Kelvin [K] \n";
    std::cout << "Enter option: ";
    std::cin >> choice;

    if (choice == 'F'|| choice == 'f' ) {
        std::cout << "----------------------------------\n";
        std::cout << "Chosen Kelvin -> Fahrenheit \n";
        std::cout << "Enter Temperature in Kelvin: ";
        std::cin >> Kelv;
        float sum = KelvtoFahr (Kelv, Fahr);

    } else if (choice == 'K'|| choice == 'k' ) {
        std::cout << "----------------------------------\n";
        std::cout << "Chosen Fahrenheit -> Kelvin \n";
        std::cout << "Enter Temperature in Fahrenheit: ";
        std::cin >> Fahr;
        float sum = FahrtoKelv (Kelv, Fahr);
```

```
30      } else {
31          std::cout << "Error";
32      }
33
34      return 0;
35  }
36
37  float KelvtoFahr (float Kelv, float Fahr){
38      Fahr = Kelv - 273.15;
39      std::cout << Fahr;
40      return Fahr;
41
42  }
43
44  float FahrtoKelv (float Kelv, float Fahr){
45      Kelv = Fahr + 273.15;
46      std::cout << Kelv;
47      return Kelv;
48
49  }
```

```
Welcome to temperature calculator
----------------------------------
Kelvin -> Fahrenheit [F]
Fahrenheit -> Kelvin [K]
Enter option: k
----------------------------------
Chosen Fahrenheit -> Kelvin
Enter Temperature in Fahrenheit: 324
597.15

=== Code Execution Successful ===
```

3. Create a C++ program that has a function that will calculate the distance between two points.
   - I had to look up for the formula to calculate the distance between the two points before I started making the function.

```cpp
#include <iostream>
#include <cmath>

float distance (int a, int b, int c, int d);
int a, b, c, d;
int main() {

    std::cout << "Input 1st point: ";
    std::cin >> a >> b;
    std::cout << "Input 2nd point: ";
    std::cin >> c >> d;
    float dist = distance (a, b, c, d);
    std::cout << "THe distance between the two points is " << dist;

    return 0;
}
float distance (int a, int b, int c, int d){
    double First = pow((b - a),2);
    double Second = pow((d - c),2);
    return sqrt(First + Second);
}

```

```
Input 1st point: 34
43
Input 2nd point: 46
89
THe distance between the two points is 43.9318

=== Code Execution Successful ===
```

4. Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

- This part of the ILO was the most difficult part as I wasn't as well versed in using classes in C++ so I had a hard time understanding the logic in the program. This part took me the longest and had me asking AI on how certain logic works from ILOB and how it should interact with the new functions created.

```cpp
#include <iostream>
#include <cmath>
#include <string>

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    bool validateTriangle() const;
    // New required function declarations
    double area(double sideA, double sideB, double sideC);
    double perimeter(double sideA, double sideB, double sideC);
    std::string triangleType() const;
};

// Constructor
Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}
```

```cpp
void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

bool Triangle::validateTriangle() const {
    return (totalAngle == 180 && angleA > 0 && angleB > 0 && angleC > 0);
}

double Triangle::area(double sideA, double sideB, double sideC) {
    double s = (sideA + sideB + sideC) / 2.0; // Semi-perimeter
    return std::sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}

double Triangle::perimeter(double sideA, double sideB, double sideC) {
    return sideA + sideB + sideC;
}

std::string Triangle::triangleType() const {
    if (!validateTriangle()) {
        return "Invalid Triangle";
    }

    if (angleA == 90 || angleB == 90 || angleC == 90) {
        return "Right-angled (Others)";
    }

    else if (angleA > 90 || angleB > 90 || angleC > 90) {
        return "Obtuse-angled";
    }
    else {
        return "Acute-angled";
    }
}
```

```cpp
int main() {
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        std::cout << "The shape is a valid triangle.\n";

        double a = 7.0;
        double b = 5.5;
        double c = 10.3;

        std::cout << "Perimeter: " << set1.perimeter(a, b, c) << "\n";
        std::cout << "Area: " << set1.area(a, b, c) << "\n";
        std::cout << "Classification: " << set1.triangleType() << "\n";

    } else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }

    return 0;
}
```

```
The shape is a valid triangle.
Perimeter: 22.8
Area: 18.0427
Classification: Obtuse-angled


=== Code Execution Successful ===
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **To ask for clarification of certain concepts and logics**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Extent of Use: **I used AI to help be refreshed after a few months of no coding from summer, it was a huge help by providing me with the information I needed to accomplish the work. Though it did not replace my actual work as I only used it as an aid.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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